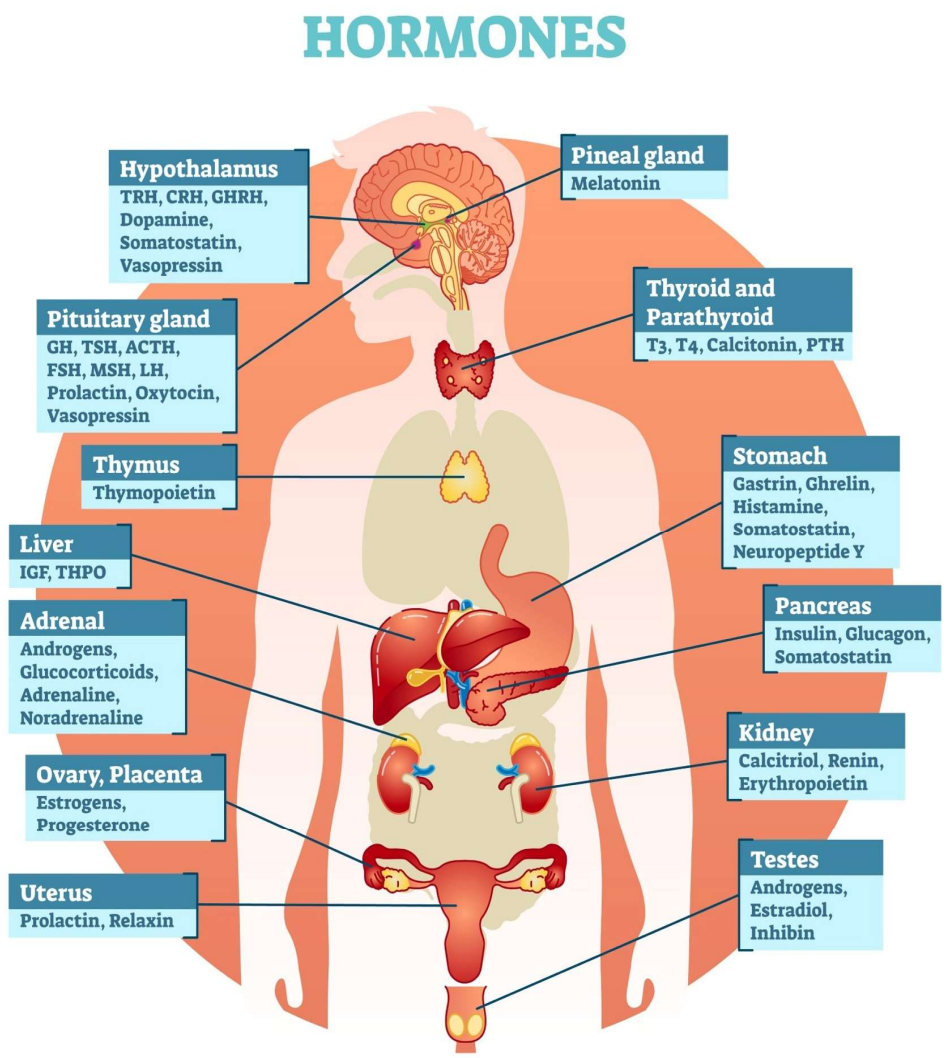
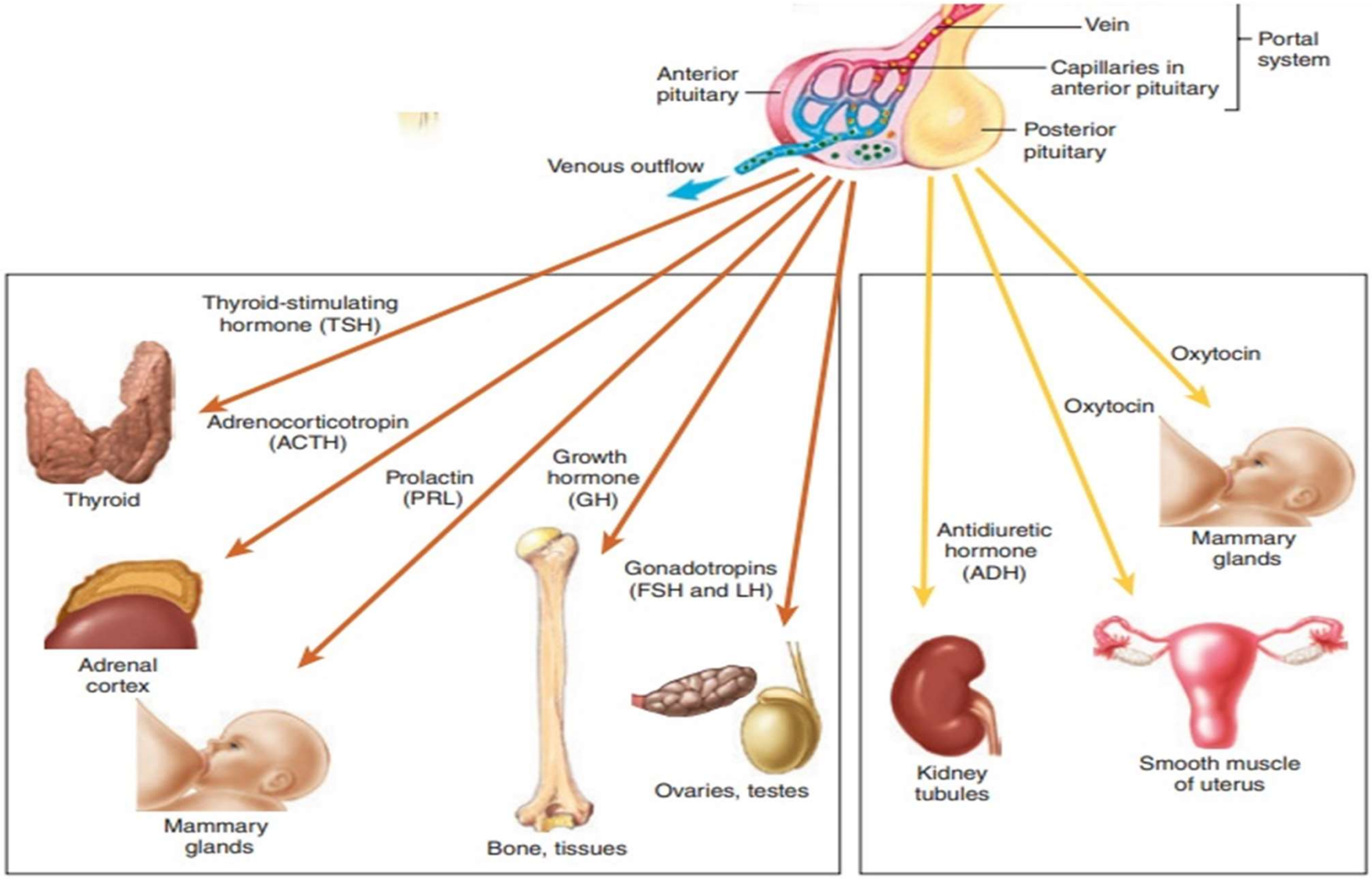
Physiology of Endocrine System (Endocrinology)

Lec2

By

staff Members of Physiology Department

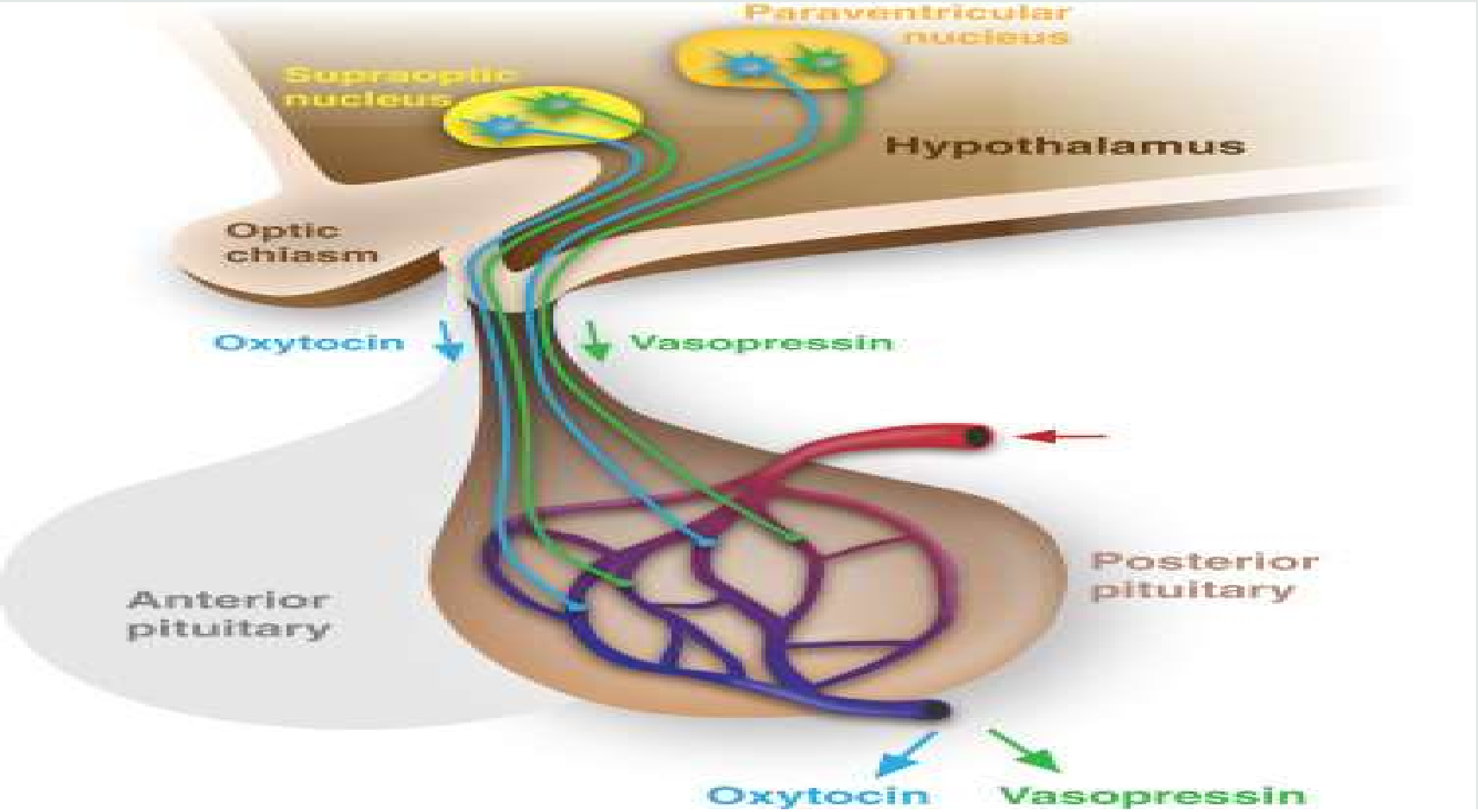




The pituitary gland.

Posterior pituitary (neurohypophysis)

- **The posterior pituitary is the back lobe of your pituitary gland, which is a small, pea-sized gland located at the base of your brain below your hypothalamus. It secretes two peptides oxytocin and Vasopressin (antidiuretic hormone, ADH) which are manufactured in the cell bodies of hypothalamic neurones.**
- **These hormones are packaged in vesicles and transported along the axons into the posterior pituitary itself to be stored in the axon terminals.**


Paraventricular nucleus
Supraoptic nucleus
Hypothalamus
Optic chiasm
Oxytocin
Vasopressin
Anterior pituitary
Posterior pituitary
Oxytocin
Vasopressin

Appropriate sensory stimuli lead to activation of the hypothalamic cells and the release of the hormones from the posterior pituitary into the bloodstream.

Neurocrine control is the process where specialized neurons in the hypothalamus, manufacture hormones, store them in vesicles (Oxytocin & ADH), and release them into the bloodstream not at a traditional synapse, so, they acting like an endocrine gland. This adaptation allows the central nervous system to directly control endocrine glands.

1- Oxytocin

is produced by cells in the paraventricular and supraoptic nuclei of the hypothalamus.

These are reflexly activated by sensory inputs from mechanoreceptors in the breast during suckling, and oxytocin is released from the posterior pituitary in response.

Functions of oxytocin include:

1- It stimulates uterine contraction during parturition and during estrus when the uterus is sensitized by estrogen.

2- Stimulates the ejection of milk through contraction of the myoepithelial cells surrounding the ducts resulting in let down of milk.

3- Oxytocin causes release of prolactin from the pituitary.

4- In laying hens, arginine vasotocin (AVT) controls oviposition via increasing in intra uterine pressure and uterine contractions by acting as an oxytocic agent, often causing premature laying upon injection.

5- It cause ovulation, Copulation causes release of oxytocin in many species which is another example of neuron-endocrine reflex.

6- It cause Luteolysis of corpus luteum.

7- In males, oxytocin stimulate transport of sperms in seminiferous tubules.

N.B) neuroendocrine reflex is a physiological mechanism where a sensory nerve stimulus is converted into a hormone release, allowing the nervous system to directly control endocrine functions.

Exs:

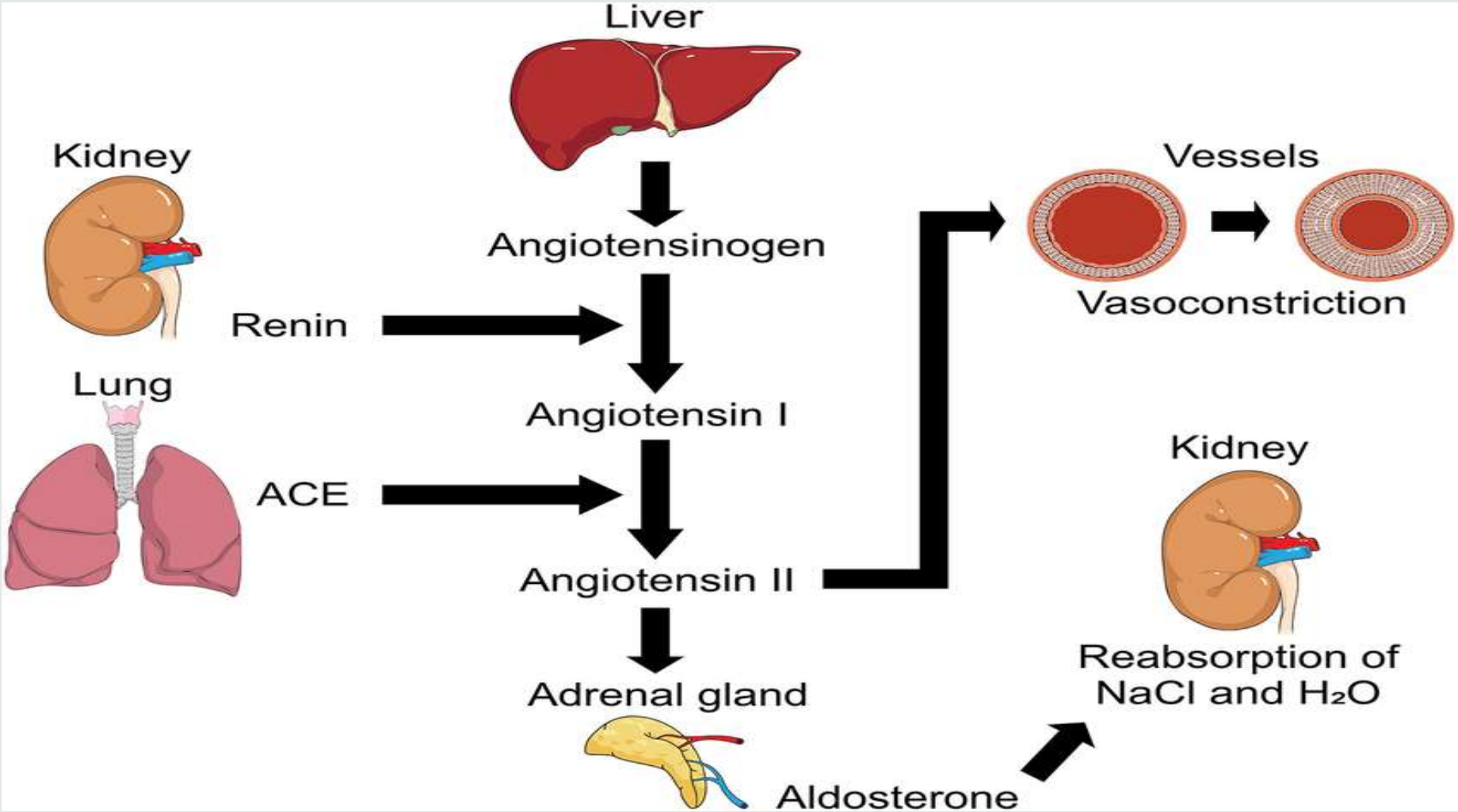
- **Milk Ejection Reflex: Stimulation of the teat/nipple activates neural signals that cause the hypothalamus to trigger oxytocin release, facilitating milk flow.**
- **Ferguson Reflex: Cervical pressure during labor sends signals to the brain, releasing oxytocin to increase uterine contractions, which is a key positive feedback loop.**
- **Stress Response: Sensory inputs (e.g., fear) trigger the hypothalamus-pituitary-adrenal (HPA) axis, causing the release of adrenaline and cortisol.**

2-Antidiuretic hormone (ADH)

- is also produced in the paraventricular and supraoptic nuclei of the hypothalamus.
- Sensory inputs, both from **local osmoreceptors** and from **stretch receptors** in the cardiovascular system (baroreceptors in aortic arch and carotid sinus),
- stimulate these cells whenever the osmolarity of the extracellular fluid rises as in case of ingestion of excess sodium chloride and in cases of dehydration), or if blood volume or blood pressure falls.

- The wall of the afferent glomerular arteriole in the kidney nephrons has Juxta glomerular cells which are sensitive to drop in the blood flow to the kidney.
- Once stimulated by decreased blood flow to the kidney these Juxta cell secrete Renin enzyme which sets in motion the rennin angiotensin mechanism which stimulate secretion of ADH and aldosterone H.

  (Explained in detail in Aldosterone H.).


Liver
Kidney
Angiotensinogen
Renin
Lung
Angiotensin I
ACE
Angiotensin II
Adrenal gland
Aldosterone
Vessels
Vasoconstriction
Kidney
Reabsorption of
NaCl and H₂O

The resulting increase in ADH:

Promotes water reabsorption from the collecting ducts of the kidney. So, tends to reduce the osmolarity and expand the volume of the extracellular fluids by promoting the production of a small volume of concentrated urine.

At the same time, peripheral resistance is increased through arteriolar constriction, which also helps to maintain arterial pressure (vasopressin effect).

Pituitary intermediate lobe

- It secrete Intermedin H, (Melanocyte –dispersing H.) sometimes called melanin dispersing H. which upon injection in a frog causes darkening of the skin.
- The hypothalamus secrete two Hs. which regulate the secretion of MSH which are MSH inhibiting (MSH-IH) and MSH RH.
- MSH-IH is more powerful than MSH-RH. That is why destruction of the hypothalamus (as following frog pithing) and consequently abolishing MSH-IH secretion result in excessive secretion of MSH-RH which stimulate release of MSH which causes dispersion of melanin pigment in the cell cytoplasm and darkens the skin.

Prolactin H

-Secretion is mainly controlled by a prolactin-inhibiting hormone (PIH) which is called, dopamine released from hypothalamus acts on D2 receptors on lactotroph cells to inhibit prolactin release.

-Stimuli which reduce PIH release from the hypothalamus raise prolactin levels.

-This occurs during pregnancy, favoring development of the mammary gland ready for lactation, and as part of the suckling reflex during suckling.

-The resultant prolactin peaks stimulate milk production (galactogenesis).

Other effects include:-

1- It has an anti-gonadotrophic action. Hyperprolactinemia causes decreased gonad function in males and females.

3-It inhibit ovulation in heavy lactating cows, the act of suckling causes release of prolactin which has luteotrophic effect that maintains the corpus luteum.

4- In birds prolactin stimulate the formation of crop milk in males and females.

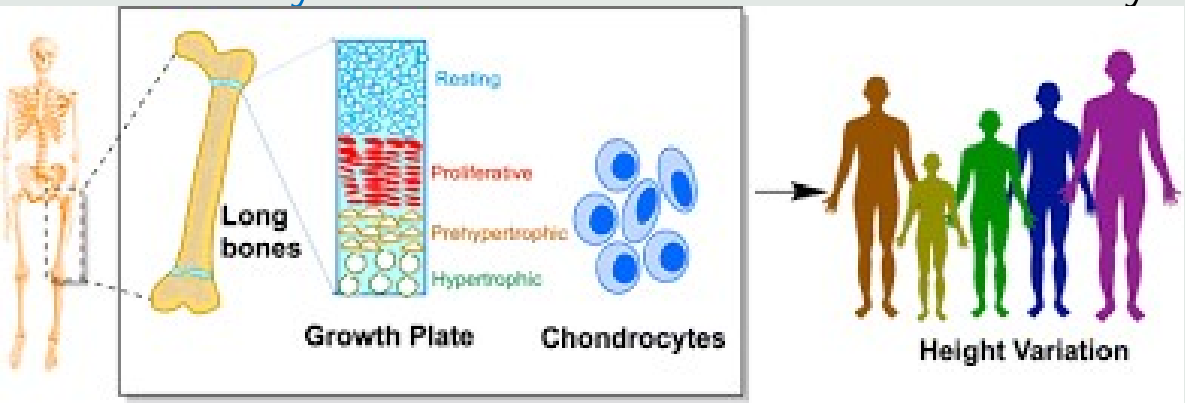
5- Responsible for parental care in fish and osmoregulation in freshwater fish (promotes ion retention and reduces osmotic water uptake, crucial for survival in low-salinity environments).

Growth hormone (GH) = somatotrophin

- is controlled by two antagonistic hypothalamic hormones

Growth hormone-releasing hormone (GH RH) is believed to be more important than the growth hormone-inhibiting hormone (GHIH) =Somatostatin.

Growth hormone is a major anabolic hormone which stimulates cell division and growth in both bony and soft tissues around the body.



- **These effects are probably indirect, being mediated by growth promoters, known as Somatomedins, produced by the liver in response to GH stimulation.**

- **Growth hormone is particularly important in young animals, and even after the epiphyses have fused, GH continues to have important metabolic functions throughout life.**

Effects of Growth H.:-

1- It decreases the excretion of nitrogenous compounds and retained nitrogen is built up into tissue protein (+ nitrogen balance) which is growth promoting effect.

2- In the post absorptive state, it limit the breakdown of body proteins for glucose production and favor the utilization of fats for energy production (GH is ketogenic).

GH is hyperglycemic.

3- GH decreases glucose usage and oxidation by cells .

During fasting GH inhibit the use of glucose by all but nerve cells (anti insulin effect).

It has diabetogenic effect Frequent injection of GH causes damage of B-cells of islets of Langerhans and the animal may die due to diabetes mellitus.

This case is called pituitary diabetes.

4- GH increases the retention of calcium and phosphate. Calcium absorption from the intestine is increased and serum phosphate level rises.

Both the soft and osseous tissues of the body increase.

5- Long bones continue to grow till closure of the epiphyseal lines at puberty under effect of gonadal hormones.

6 - It has lactogenic effect (acting on mammary gland prolactin receptors, stimulating milk synthesis, alveolar development, and increasing yield. It enhances mammary metabolism and can increase lactose, fat, and protein production, acting as both a somatogenic and lactogenic hormone)

Administration of GH to dairy cows increases milk yield by approximately 32%, with concomitant increases in milk protein, fat, and lactose yield.

Control of GH release:-

1. **Hypothalamic control. Decreased output of somatostatin results in more secretion of GH.**
2. **Hypoglycemia act as a potent stimulus of GH secretion.**
3. **Stress and sleep stimulate secretion of GH RH.**
4. **GH. Feeds back to inhibit its own secretion.**

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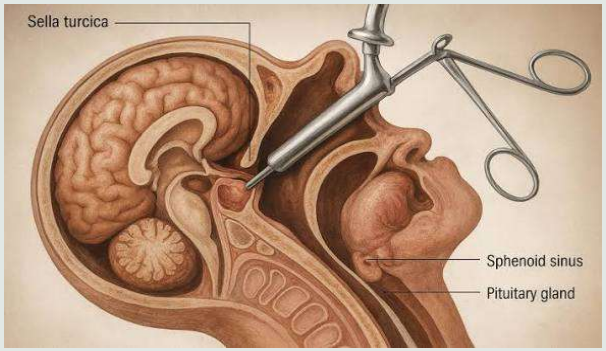
Deficient pituitary function:



(e.g :Hypophysectomy)

- Deficiencies resulting from damage the entire gland, e.g. because of a local tumor or its surgical removal, may involve all the pituitary hormones.
- Alternatively, there may be a deficit in a single hormone caused by a defect in the relevant pituitary cells or in the hypothalamic cells which regulate their function.

The main problems which can result in:

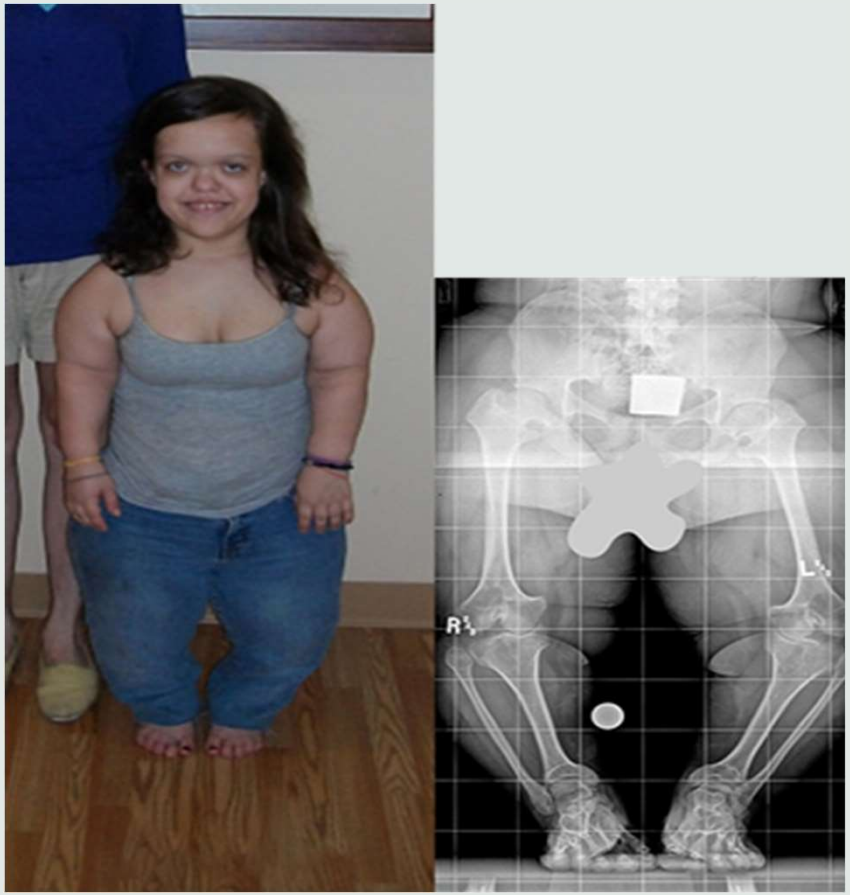
1- Corticosteroid deficiency caused by lack of ACTH.

2- Thyroid deficiency caused by lack of TSH.

3- Failure of sexual function caused by lack of FSH/LH.

4- Dwarfism caused by lack of GH in young may arise as a congenital defect and can be treated with GH supplements.

Dwarfism



5- Diabetes insipidus caused by lack of ADH

this results in an inability to concentrate urine

so the patient passes 8–10 L of dilute urine per day and has to drink a commensurate volume.

Excess secretion:-

The following conditions resulting from over secretion of a single pituitary hormone are observed:-

1- Corticosteroid excess caused by elevated ACTH

2-Impaired reproductive function caused by elevated prolactin. Hyperprolactinaemia which is now recognized as an important cause of infertility. This seems to be caused by prolactin's ability to block GnRH production by the hypothalamus.

4- Fluid retention and low plasma osmolarity caused by elevated ADH.

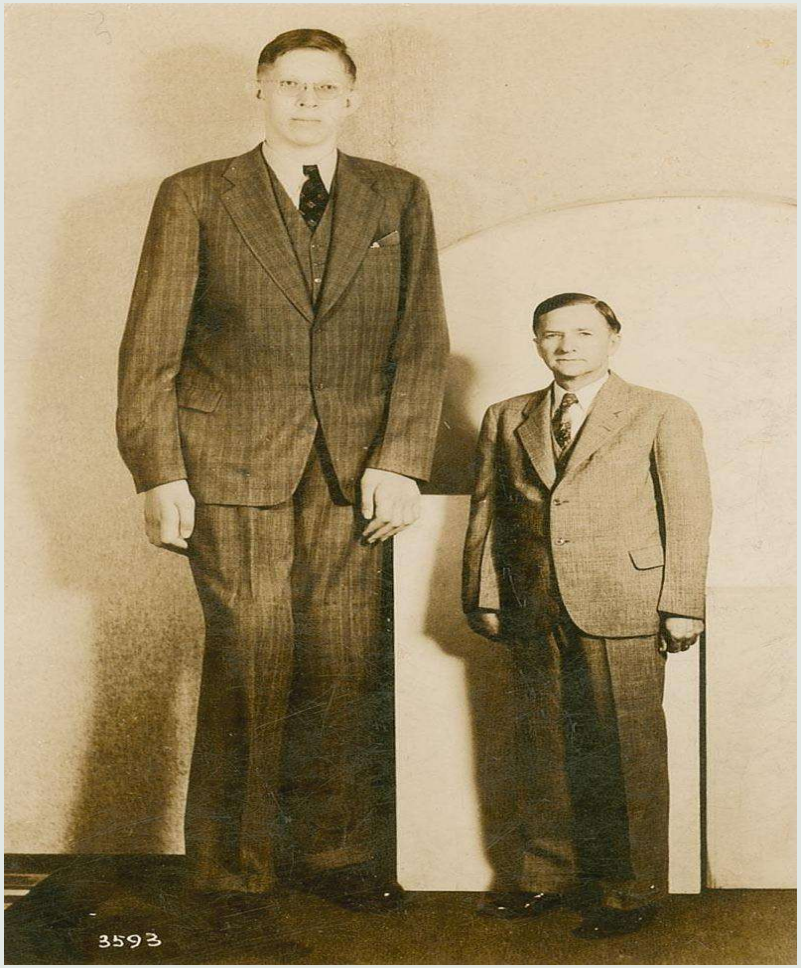
3- Abnormal growth caused by elevated GH.

1)This stimulates long bone growth in young causing Gigantism

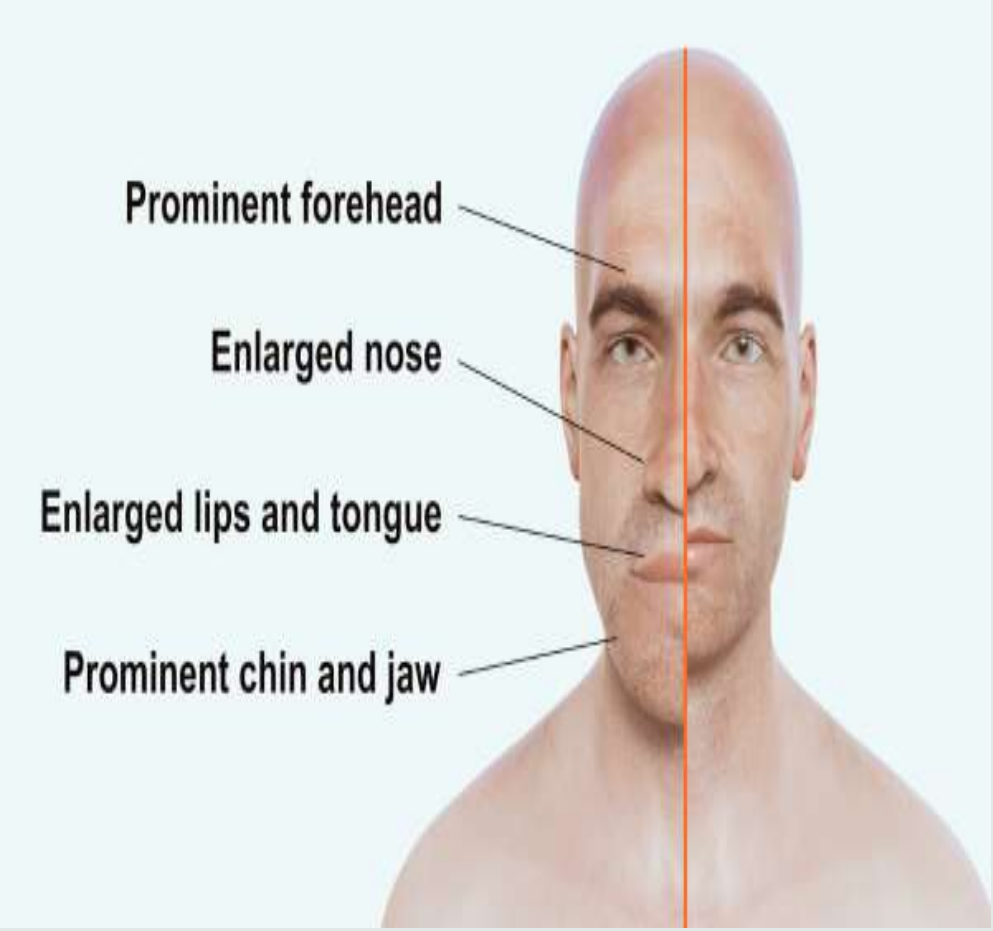
2)If GH levels become raised in adult life, after epiphyseal fusion, there is no increase in height, but soft tissue and body organ growth are stimulated. This is known as Acromegaly.

(Acromegaly a persistent excess of GH may eventually lead to elevated blood glucose levels, i.e. diabetes mellitus results.)

Gigantism



Acromegaly


Prominent forehead
Enlarged nose
Enlarged lips and tongue
Prominent chin and jaw

Acromegaly

Insulin-like growth factor 1 (IGF-1)= somatomedin C

-is a protein H similar in molecular structure to insulin.

-Consists of 70 amino acids .

-It plays an important role in young age growth and continues to have anabolic effects in adults.

IGF-1 is produced primarily by the liver as an endocrine hormone as well as in target tissues in a paracrine/autocrine fashion.

Production is stimulated by growth hormone (GH) and can be retarded by undernutrition, growth hormone insensitivity, lack of growth hormone receptors.

IGF-1 is produced throughout life. The highest rates of IGF-1 production occur during the pubertal growth spurt.

The lowest levels occur in infancy and old age.

Mechanism of action:-

IGF-1 is a primary mediator of the effects of growth hormone (GH).

Growth hormone is made in the anterior pituitary gland, is released into the blood stream, and then stimulates the liver to produce IGF-1.

IGF-1 then stimulates systemic body growth ,and has growth-promoting effects on almost every cell in the body, especially skeletal muscle, cartilage, bone, liver, kidney, nerves, skin, hematopoietic cell, and lungs.

- **In addition to the insulin-like effects, IGF-1 can also regulate cell growth and development, especially in nerve cells, as well as cellular DNA synthesis.**
- **Body builders inject IGF-1 into individual skeletal muscles to increase its size.**
- **Protein intake increases IGF-1 levels , independent of total calorie consumption.**

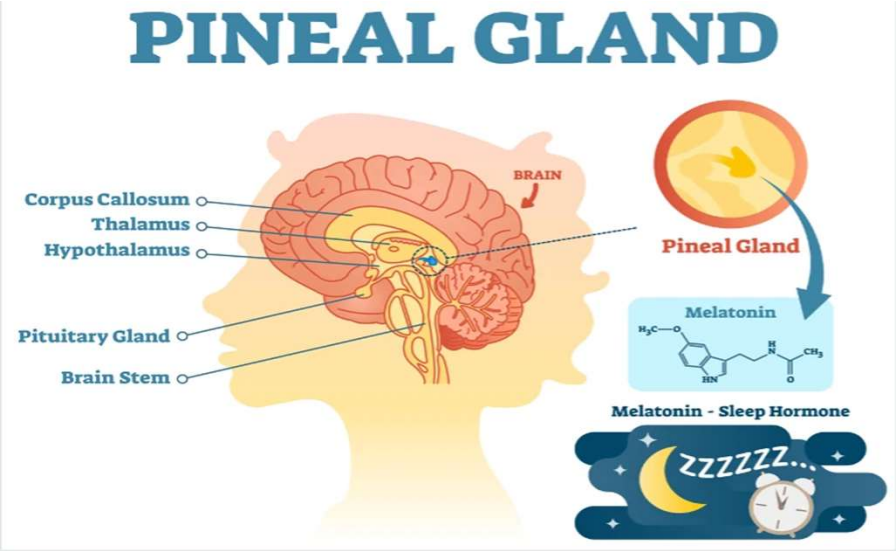
Factors that are known to cause variation in the levels of growth hormone (GH) and IGF-1 in the circulation include:

(insulin levels, genetic make-up, the time of day, age, sex, exercise status, stress levels, nutrition level and body mass index (BMI), disease state, estrogen status and xenobiotic intake.)

(xenobiotic : chemical substances foreign to a biological system, including synthetic drugs, environmental pollutants, industrial chemicals, food additives, and pesticides)

- **IGF-1 activates the insulin receptor at approximately 0.1 times the potency of insulin.**
- **Part of this signaling may be via IGF1R/Insulin Receptor heterodimers .**
- **The actual potency of IGF1 in vivo is inducing phosphorylation of the insulin receptor, and hypoglycemia).**

The Pineal gland



- Lies on the roof of the third ventricle between the two thalami

it is composed of glia and secretory –type cell.

- It secrete Melatonin H. during dark hours which have inhibitory action on the gonads.
- In the frog and birds the pineal gland acts as a photoreceptor.
- It keeps the circadian cycles in tune .

Light acts via the pineal gland to control the reproductive cycles.

Melatonin in small amounts improve spermatogenesis.

Injection of melatonin H. resulted in:

1-decreased ovarian weight via acting as an antigonadotropic agent that suppresses follicle stimulation,

2-inhibit the cyclic release of LH via suppressing the stimulatory effects of gonadotropin-releasing hormone (GnRH/LHRH)

3-lengthened estrous cycle in rats (anti-gonadotropic agent).

-It has antioxidant property.

-Bovine pineal gland contains a factor that blanches pigment cells in tadpoles.

Q: Laying hens are raised under 16 h light and 8 hours darkness cycle: Explain?

The pineal gland in laying hens acts as a central pacemaker in the circadian system, controlling 24-hour rhythms in behavior and physiology, particularly through the nocturnal secretion of melatonin.

Melatonin levels peak during darkness, guiding sleep and metabolic processes.

Long (14–16 hour) light periods are used to stimulate laying, as the pineal gland's regulation of circadian rhythms impacts the reproductive cycle.

thank
you